



EduGenie

AI-Powered Educational Assistant

Ask • Learn • Grow

Project Name

EduGenie is an AI-powered educational assistant, which is mainly designed to help students learn better and make their study process easier.



PROJECT SUBMISSION

- TEAM DETAILS -



Project Name : EduGenie: Google Gemini Powered Learning Assistant



Project Type : Generative AI Project



Team Name

S.No.	Role	Name	Register No.
1	Team Lead	Aadhithyan.A	25CAA3401
2	Team Member	Ariharan.Ap	25CAA3403
3	Team Member	Arivukkarasan.A	25CAA3404
4	Team Member	Mohamed Thovfeeq.M	25CAA3421
5	Team Member	Ganesan.A	25CAA3407



Institution Details

College Name : H.H. The Rajah's College, Pudukkottai

Department : Computer science

Course / Programme : B.C.A

Academic Year : 2026-2027

Project Guide / Faculty : EduGenie



Project Description

EduGenie is a lightweight AI-powered educational assistant that uses generative AI to help learners ask questions, simplify concepts, generate quizzes, receive learning recommendations, and summarize educational content.



EduGenie

Google Gemini Powered Learning Assistant

Ask • Learn • Grow

Your AI-powered
study buddy for a
smarter tomorrow!



Project Overview

EduGenie is a lightweight AI-powered educational assistant that simplifies learning through generative AI. Built with FastAPI for the backend and a simple HTML + CSS frontend, EduGenie leverages Google Gemini models for smart, accurate and helpful responses.



Key Features



Ask Questions

Get smart, accurate answers to any subject.



Simplified Explanations

Understand complex concepts easily.



Generate Quizzes

Create practice questions for your own test.



Personalized Recommendations

Get learning paths and suggestions just for you.



Summarization Test

Turn long passages into short, easy-to-read notes.



Real-Life Scenarios

1

Ask a question

"What is the impact of water?"

2

Generate a quiz

"Create a 5-question MCQ quiz on Photosynthesis."

3

Get a learning path

"I want to learn Python. Create a step-by-step learning path with resources and practice exercises."

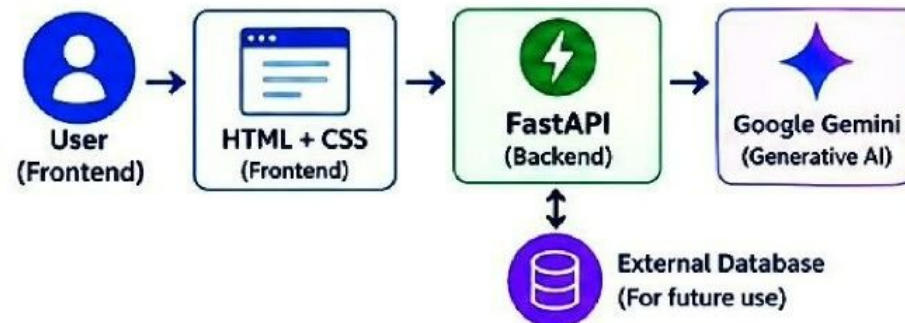


Smart Learning + Powered by Gemini

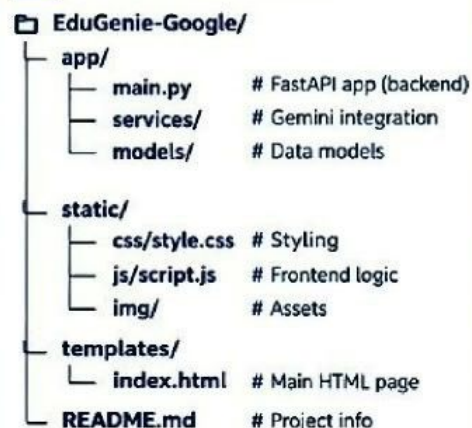
EduGenie makes education interactive, engaging and personalized.



Architecture Diagram



Project Structure



EduGenie

Learn Smarter | Build Your Future



Pre-requisites



Python 3.10+



Google Gemini API Key



Install dependencies

```
pip install -r requirements.txt
```



Setup Steps

1

Set up Gemini API key (Google Gemini) and add to .env file

2

Install required packages

3

Run the FastAPI server

```
uvicorn app.main:app --reload
```

4

Open your browser and visit

```
http://localhost:8000
```



Python Code Example

```
1 import google.generativeai as genai
2
3 # Set up Gemini API
4 genai.configure(api_key="YOUR_API_KEY")
5 # Create model
6 model = genai.GenerativeModel('gemini-1.5-pro')
7 # Generate response
8 response = model.generate_content(
9     "Explain machine learning in simple terms."
10 )
11 print(response.text)
```



Output Example

"Machine learning is a branch of AI that allows computers to learn from data, find patterns, and make predictions without being explicitly programmed."



Other Language Examples

JS JavaScript

```
const { GoogleGenerativeAI } =
  require('@google/generative-ai');

const genAI = new GoogleGenerativeAI(
  'YOUR_API_KEY');
```



```
#include <iostream>
#include <genai.h>

int main() {
    // Initialize Gemini API
    return 0;
}
```



```
import com.google.generativeai as genai;

genai.configure(apiKey="YOUR_API_KEY");
GenerativeModel model = new
  GenerativeModel('gemini-1.5-pro');
```



```
import google.generativeai as genai

genai.configure(api_key="YOUR_API_KEY")
model = genai.GenerativeModel(
    'gemini-1.5-pro'
)
print(model.generate_content("Hello"))
```



```
1 import google.generativeai as genai
2
3 def summarize_text(text: str) -> str:
4     try:
5         model = genai.GenerativeModel(
6             model_name='models/gemini-1.5-pro'
7         )
8         prompt = f"Summarize the following text\n{ text }"
9         response = model.generate_content(
10             prompt, request_options={"max_output_tokens": 200}
11         )
12         return response.text.strip()
13     except Exception as e:
14         return f"Error in Summary: {e}"
```



Output Example

"The text is about the AI field, its evolution by providing quick answers, simple explanations, quotes and personalized learning paths."



Other Language Examples



JavaScript

```
const { GoogleGenAI } = require('google-genai');
const ai = new GoogleGenAI({
  apiKey: 'YOUR_API_KEY'
});

ai.models.generateContent({
  model: 'gemini-1.5-pro',
  contents: "Summarize the text"
});
```



Java

```
import com.google.genai.Client;
import com.google.genai.model.GenerateContentRequest;
import com.google.genai.model.Content;

Client client = new Client();
GenerateContentRequest request = new
    GenerateContentRequest.Builder()
        .model("gemini-1.5-pro")
        .contents(Content.of("Summarize the text"))
        .build();

String response = client.generateContent(request);
```



C++

```
#include <iostream>
#include <google/generative-ai.h>

int main() {
    // Example API call
    return 0;
}
```



Python (API Client)

```
import google.generativeai as genai
genai.configure(api_key="YOUR_API_KEY")
model = genai.GenerativeModel("gemini-1.5-pro")
response = model.generate_content(
    "Summarize the following text"
)

print(response.text)
```

3. Features in Action



Here's how EduGenie helps you learn:



Ask a Question

What is the largest ocean in the world?

The Pacific Ocean is the largest ocean in the world.



Explain a Topic

Explain photosynthesis simply

Photosynthesis is the process by which plants make food using sunlight, carbon dioxide and water.



Generate Quiz

Python basics

1. What is Python?

- A) A snake
- B) A programming language
- C) A car
- D) A fruit



Summarize Text

Python is a high-level programming language...

Key Points:

- Easy to learn
- Versatile
- Widely used



Learning Path

I want to learn SQL from beginner to advanced

Learning Path:

1. SQL Basics
2. Intermediate Queries
3. Advanced Topics
4. Practice Projects

5. Run & Use

Follow these steps to start EduGenie on your computer:

1

Activate virtual environment

```
.\venv\Scripts\Activate.ps1
```



2

Install dependencies

```
python -m pip install -r requirements.txt
```



3

Start the server

```
python -m uvicorn main:app --reload
```





Thank You!




EduGenie

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